



- An **exponent** symbol in Excel is the “**caret**”: ^

EXAMPLE: 2^3 would be coded as: 2^3

- The **multiplication** symbol is the “**asterisk**”: *

EXAMPLE: 2×3 would be coded as: $2*3$

- The **division** symbol is the forward slash: /

EXAMPLE: $9 \div 3$ would be coded as: $9/3$

- The **addition** symbol is the **plus sign**: +

EXAMPLE: $2+3$

- The **subtraction** symbol is the **minus sign**: -

EXAMPLE: $3-2$



“Operator Precedence”

operator precedence

Operation:	Operator:	Examples:		
Exponentiation	$\wedge$	$=2\wedge3$	$=A3\wedge2$	$=A3\wedge B4$
Multiplication	$*$	$=2*3$	$=A3*2$	$=A3*B4$
Division	$/$	$=2/3$	$=A3/2$	$=A3/B4$
Addition	$+$	$=2+3$	$=A3+2$	$=A3+B4$
Subtraction	$-$	$=2-3$	$=A3-2$	$=A3-B4$

- Any **exponents** in a formula are performed first, followed by any **multiplication** or **division**, followed by any **addition** or **subtraction**
- You can override this order with **parentheses**, which also improve the **readability** of formulas:

$$= 3 * 4 \wedge 2$$

$$= (3 * 4) \wedge 2$$



Entering Formulas

Entering Formulas

- One way that we could **add up** cells **B1**, **B2**, and **B3** and have the sum appear in cell **B4** would be to enter the following formula in cell **B4**:

	A	B	C
1		10	
2		20	
3		30	
4	Total:	60	

- TIP:** rather than typing a cell address into a formula, **click on the actual cell** instead; Excel will enter that cell's **address** into the formula for you !

Ranges

- A “**range**” of cells is denoted by the first cell and the last cell in the range, separated by a colon:

	A	B
1		
2		
3		
4		

B2 : B4

	A	B	C
1			
2			

A2 : C2

	A	B	C
1			
2			
3			
4			

A2 : C4

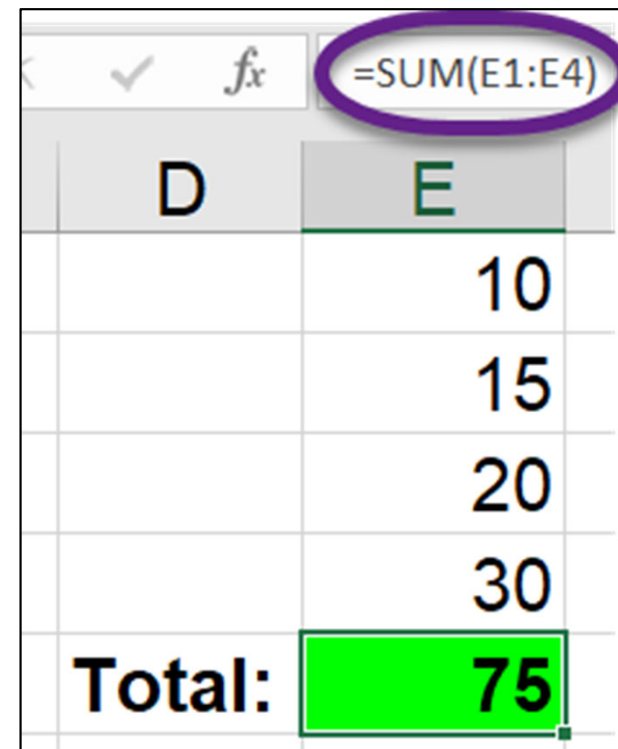
- TIP**: enter ranges by **clicking and dragging...**

Adding Up a Range

Ranges & Functions

SUM() function: the better way to **total** a row or column!

=SUM(E1:E4)



The screenshot shows an Excel spreadsheet with two columns, D and E. Column E contains the values 10, 15, 20, and 30. In column D, the cell below the header is labeled 'Total:'. The cell to the right of 'Total:' is highlighted in green and contains the value 75. A formula bar at the top right shows the formula =SUM(E1:E4) circled in purple.

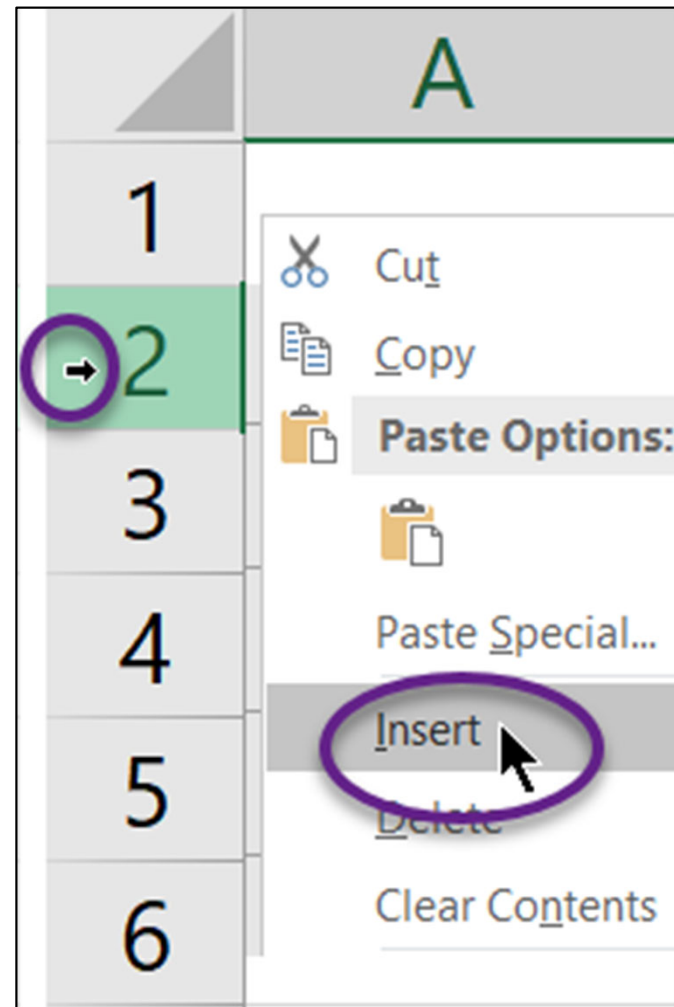
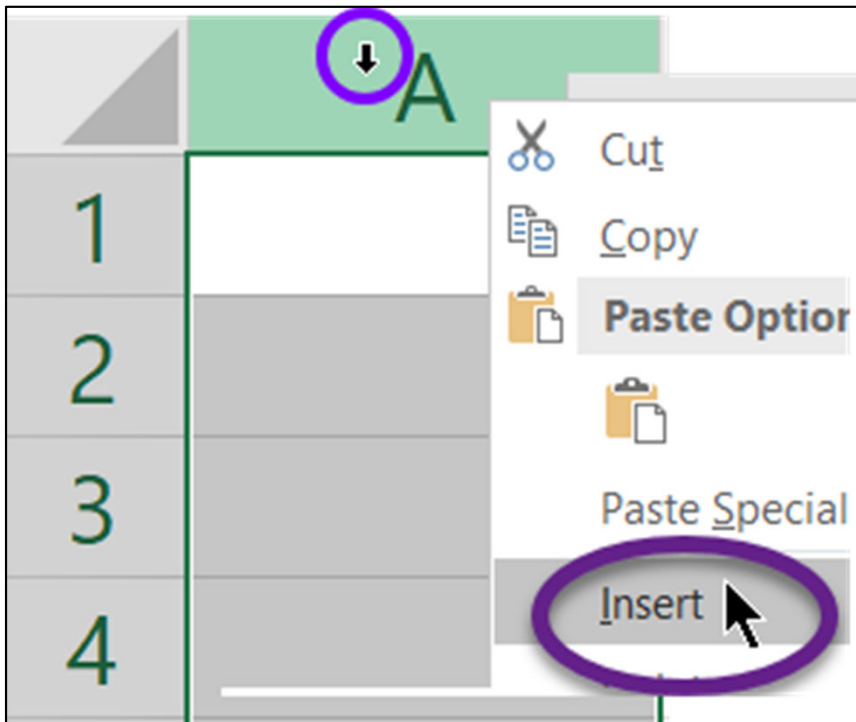
D	E
	10
	15
	20
	30
Total:	75

Why is SUM() a better way to calculate a total?

- Because the **SUM()** function can use **ranges**... and ranges will **expand** or **contract** when you add or delete a row or column!

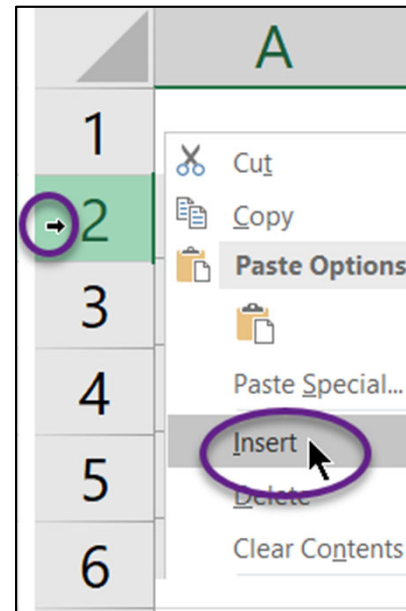
To insert or delete a row/column:

- **Right-click** on the **row/column heading** and select **Insert** or **Delete** from the menu that appears:



Example: Inserting a Row

- **Right-click** on the **row 2** heading and select **Insert**:



Before:

	A	B	C	D	E
1		10			10
2		15			15
3		20			20
4		30			30
5	Total:	75		Total:	75
6					

After:



	A	B	C	D	E
1		10			10
2					
3		15			15
4		20			20
5		30			30
6	Total:	75		Total:	75

- **Row 2 is now empty.**

Notice: Range has expanded!

- Notice that the range used in the **SUM()** function has automatically expanded with the insertion of the new **row**, from **E1:E4** to **E1:E5** ...

Before:



fx		=SUM(E1:E4)
D	E	
		10
		15
		20
		30
Total:		75

After:



fx		=SUM(E1:E5)
D	E	
		10
		15
		20
		30
Total:		75

- If we add a new value in the new row
 - for example **12** in cell **E2** – then the correct sum is calculated and appears in cell **E10** ...

Before:

✓	<i>fx</i>	=SUM(E1:E5)
D	E	
		10
		15
		20
		30
Total:		75

After:

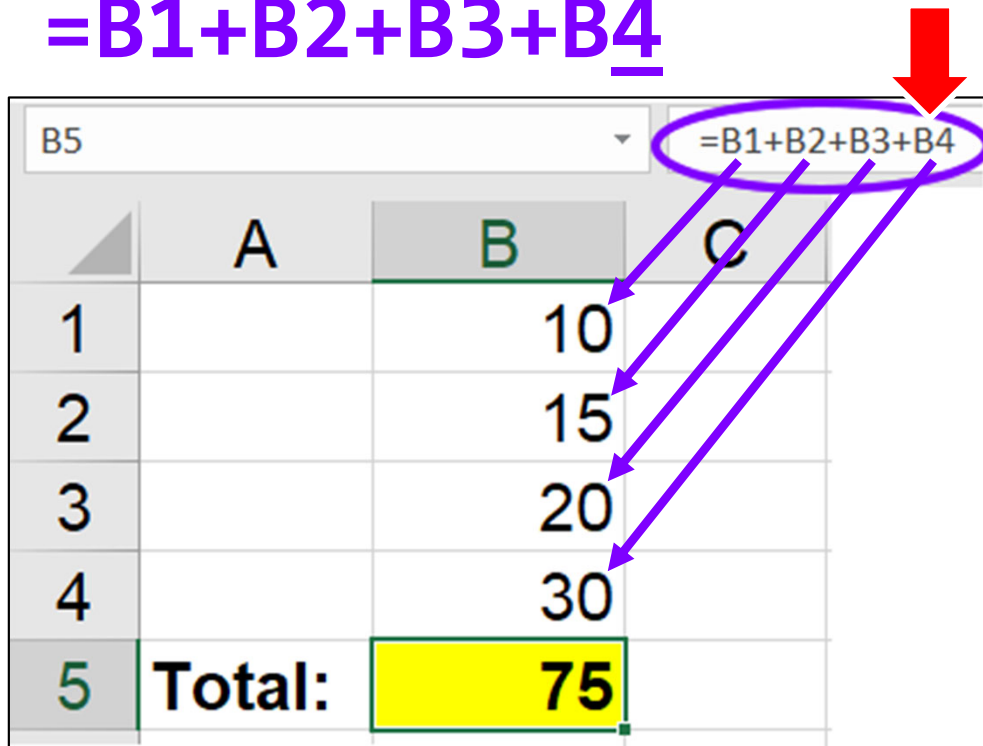
✓	<i>fx</i>	=SUM(E1:E5)
D	E	
		10
		12
		15
		20
		30
Total:		87

On the other hand...

Notice the **change in the formula** in cell **B5** when we inserted the **new row**...

from:

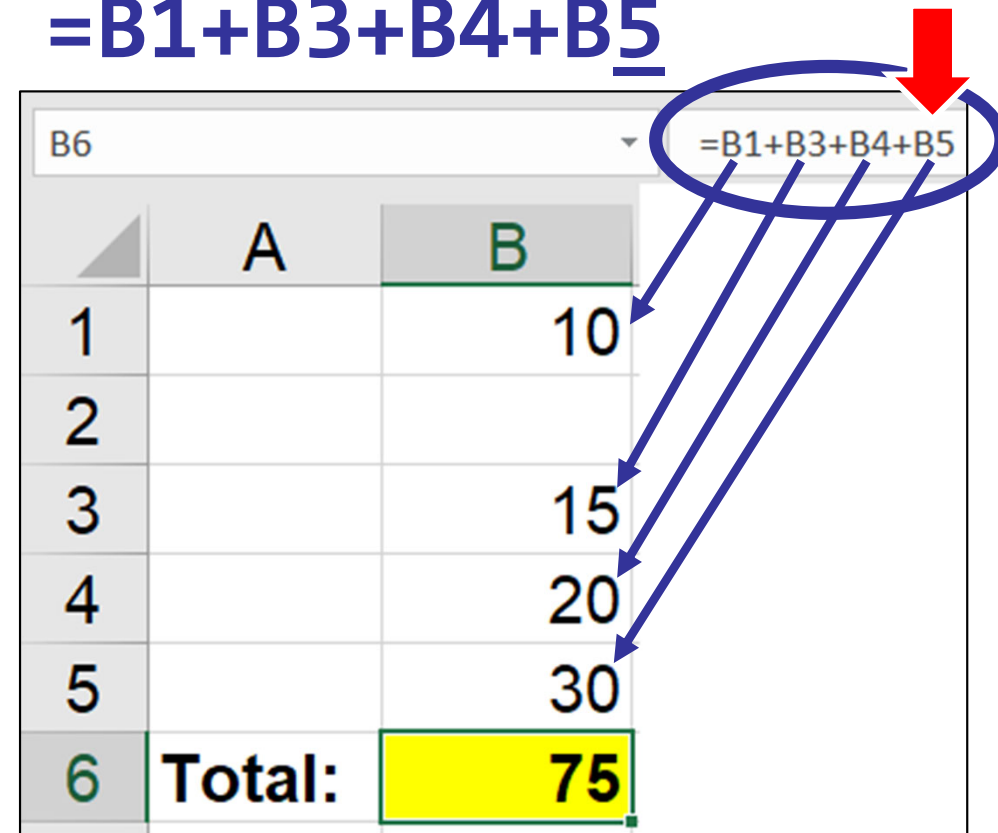
$=B1+B2+B3+B4$



	A	B	C
1		10	
2		15	
3		20	
4		30	
5	Total:	75	

to:

$=B1+B3+B4+B5$

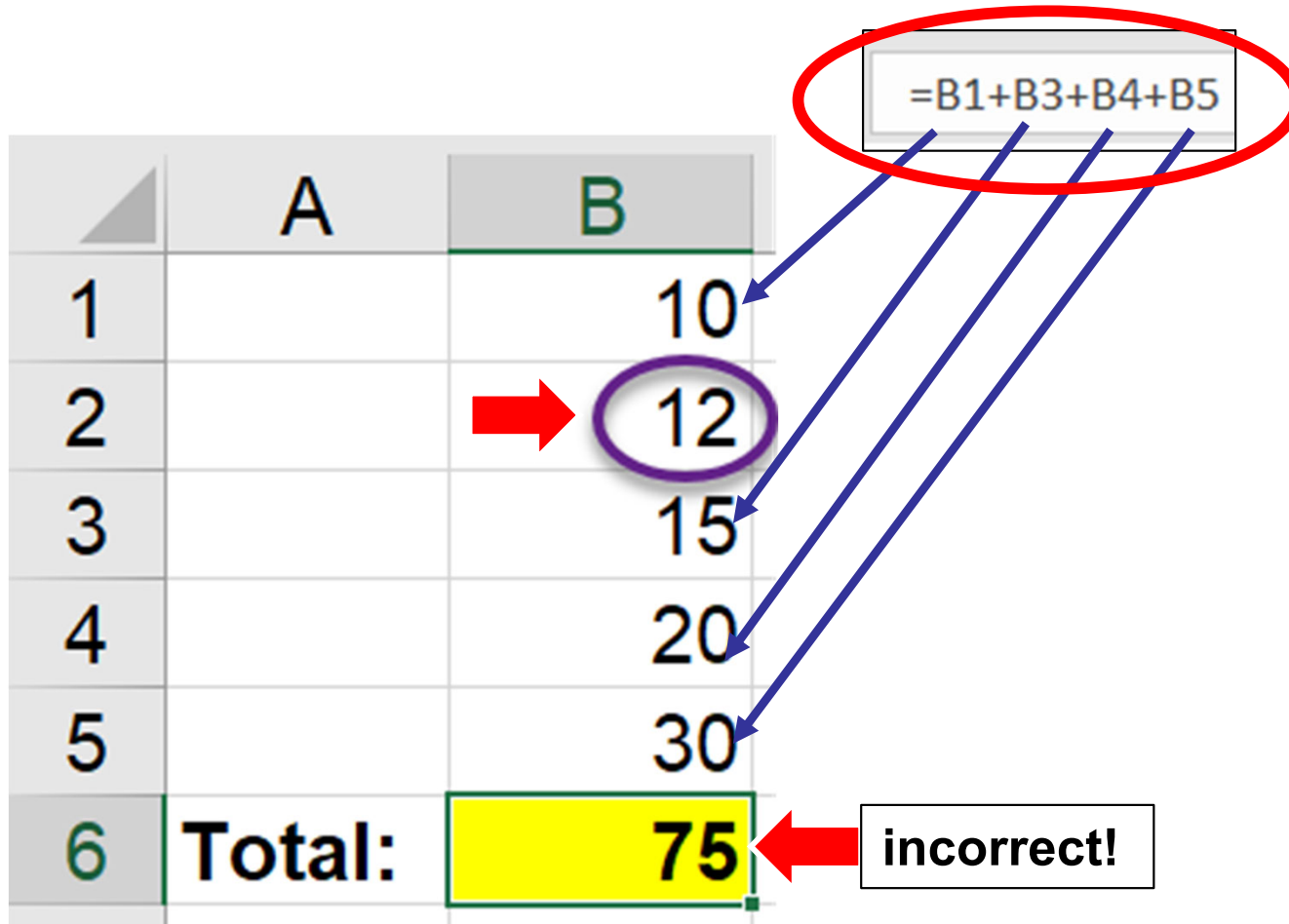


	A	B
1		10
2		
3		15
4		20
5		30
6	Total:	75

- Thus, cell **B2** is no longer included in the **total**!

Erroneous Total

Thus, even if we add a new value in cell **B2** ...



	A	B
1		10
2		12
3		15
4		20
5		30
6	Total:	75

Formula bar: `=B1+B3+B4+B5`

incorrect!

...this value in cell **B2** is **not included** in the **total**.

Thus, **B6**, the cell labeled as the total, is **not correct!**

Correct and Incorrect Totals



As we have seen, **inserting** a new **row** or **column** and **new values** is a good illustration of the advantage of using **ranges**:

	A	B	C	D	E
1		10			10
2		12			12
3		15			15
4		20			20
5		30			30
6	Total:	75		Total:	87

The table illustrates the advantage of using ranges for calculations. It shows two scenarios for calculating totals:

- Incorrect:** A red arrow points from the "incorrect!" label to the yellow cell containing the total value of 75. This result is based on a range that does not include the newly inserted row (row 5).
- Correct:** A red arrow points from the "correct!" label to the green cell containing the total value of 87. This result is based on a range that includes all data, including the newly inserted row (row 5).

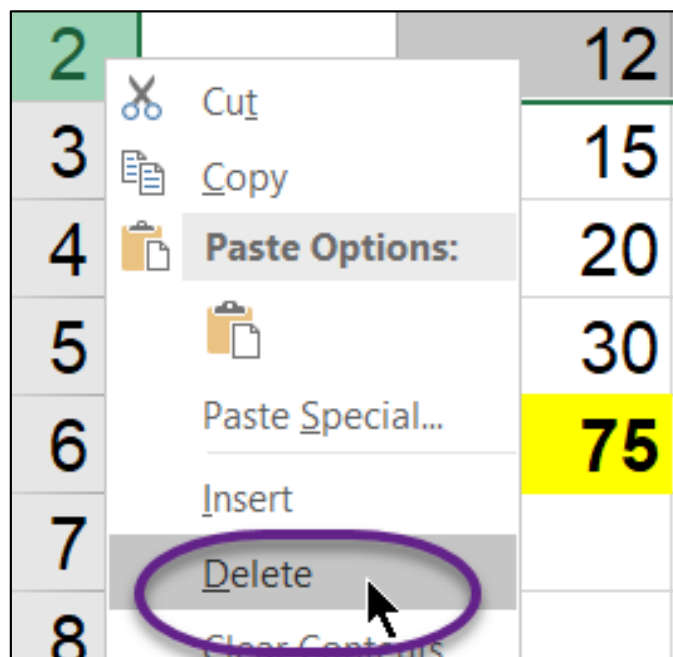
Deleting a Row



- **Right-click** on row 2

	A	B	C	D	E
1		10			10
2		12			12
3		15			15
4		20			20
5		30			30
6	Total:	75		Total:	87

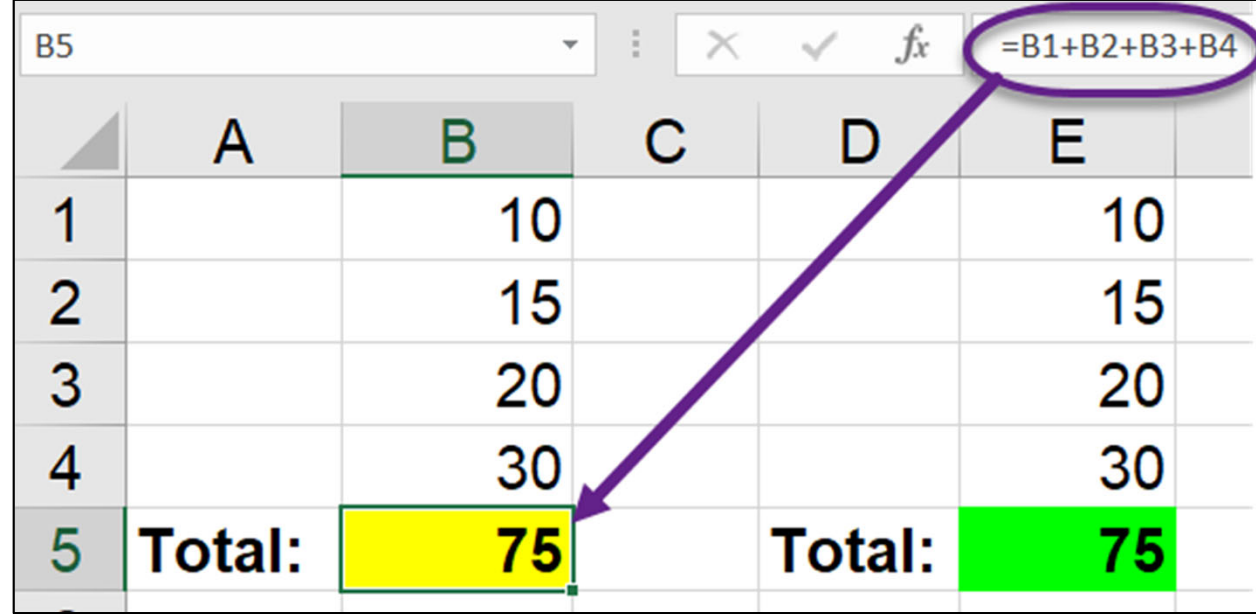
- Select **Delete** from the menu that appears:



Result of Deleting a Row

The formula in cell **B5** returns to:

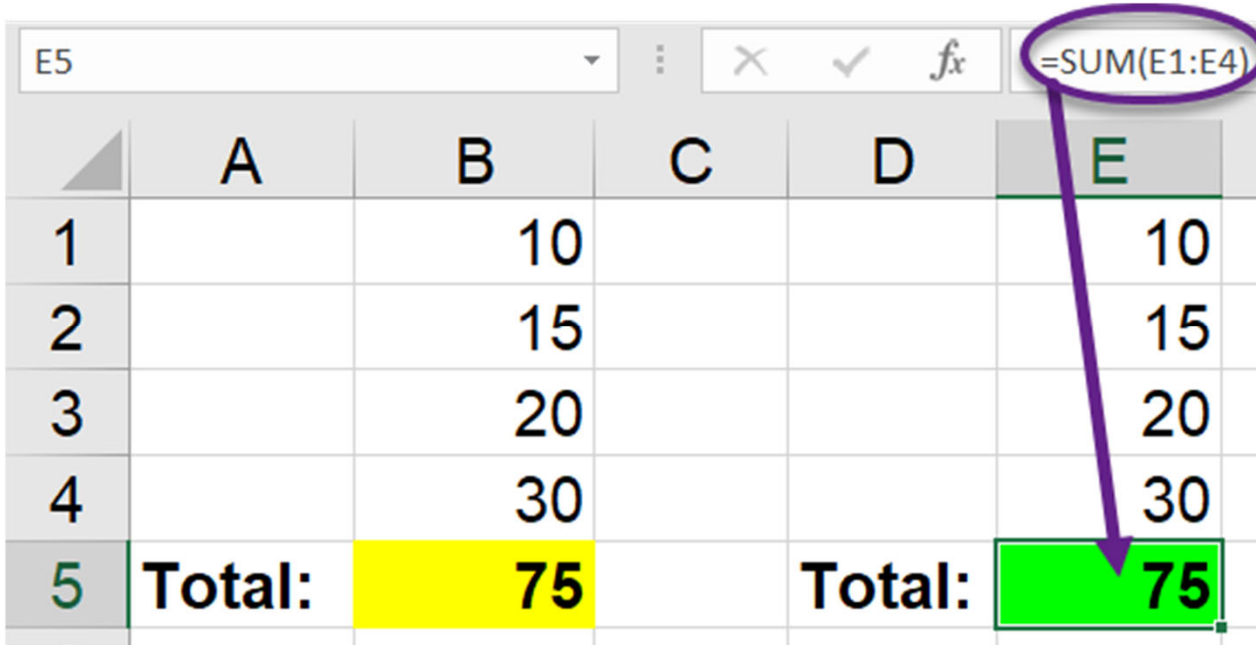
=B1+B2+B3+B4



	A	B	C	D	E
1		10			10
2		15			15
3		20			20
4		30			30
5	Total:	75		Total:	75

The formula in cell **E5** returns to:

=SUM(E1:E4)



	A	B	C	D	E
1		10			10
2		15			15
3		20			20
4		30			30
5	Total:	75		Total:	75

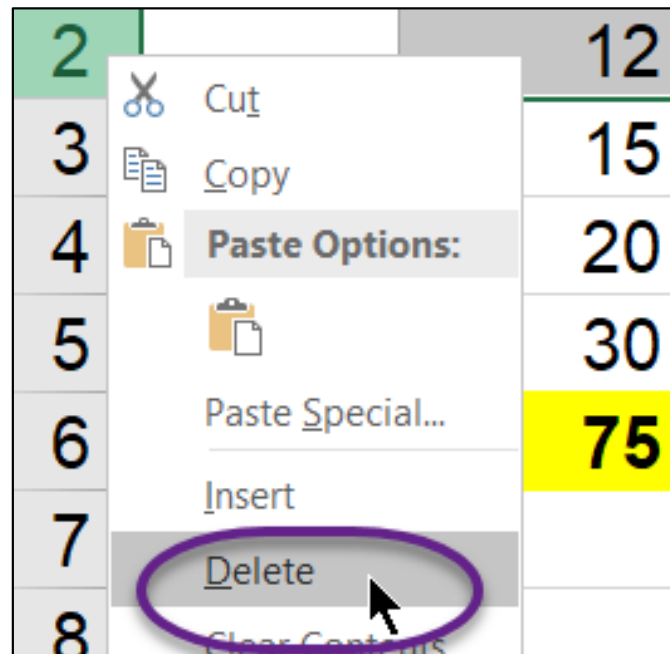
Deleting a Row



- **Right-click** on row 2

	A	B	C	D	E
1		10			10
2		12			12
3		15			15
4		20			20
5		30			30
6	Total:	75		Total:	87

- Select **Delete** from the menu that appears:



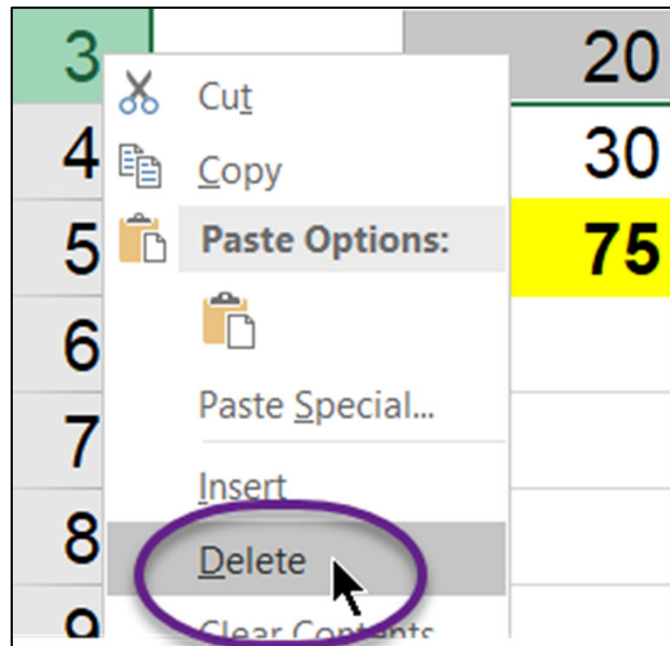
Deleting Another Row



- **Right-click** on row 3

	A	B	C	D	E
1		10			10
2		15			15
3		20			20
4		30			30
5	Total:	75		Total:	75

- Select **Delete** from the menu that appears:





Result of Deleting Another Row

Ranges & Functions

The range in the formula in cell E5 has automatically changed from **=SUM(E1:E4)** to **=SUM(E1:E3)**

	A	B	C	D	E
1		10			10
2		15			15
3		30			30
4	Total:	#REF!		Total:	55

Thus, the **total** in cell E5 is still correct:

D	E
	10
	15
	30
Total:	55



Result of Deleting Another Row

Ranges & Functions

However, the formula in cell **B5** has automatically changed from **=B1+B2+B3+B4** to **=B1+B2+#REF+B3**

	A	B
1		10
2		15
3		30
4	Total:	#REF!

#REF is an Excel error message.

This error message has appeared because a cell used in the formula has been deleted: cell **B3**

Undo (Ctrl-Z)

Ranges & Functions

- To undo this error, click Excel's **Undo** button...



...or, press **Ctrl-Z** on your keyboard.

- Afterward, the **deleted row** should now be restored:

	A	B	C	D	E
1		10			10
2		15			15
3		20			20
4		30			30
5	Total:	75		Total:	75